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Dear Sir/Madam,

I am writing to propose my perspective paper, “Engineering the atomic future: The case for responsible modular reactors”, for publication in the journal *Science*.

This perspective paper explores the environmental implications of small modular reactors deployment, particularly from high fuel enrichment and complex waste management. As a key contributor to a decarbonised energy system, addressing these challenges will be essential to maintain the sustainability and public perception of nuclear power.

Thank you for your time and consideration.

Sincerely,
Author

NUCLEAR

Engineering the atomic future: The case for responsible modular reactors

ABSTRACT

SMR has an important role to play in a decarbonised energy system. Increasing use of HALEU and complex waste characteristics of SMR induce lifecycle emissions, problematic waste streams, and other environmental implications. Research, governmental partnerships and HALEU fuel cycle and waste management infrastructure are essential for the sustainable deployment of SMRs.

1. Introduction

In the pursuit of a low-carbon, baseload energy source, nuclear power is a viable alternative. The challenge of achieving net-zero has reasserted its importance in global energy and climate discussions as a potential key contributor to a decarbonised energy system. Global installed nuclear capacities are projected to reach 1,160 GWe by 2050 in a net-zero scenario, almost triple the current capacity (1).

Small modular reactors (SMRs) are hailed to overcome long lead times and capital cost overruns plagued by large nuclear reactors (LRs) (2). SMRs operate on a smaller scale, up to 300 MWe, compared to LRs in the thousands of MWe. They could play a huge role in decarbonising a variety of industries, such as low-carbon electricity generation, water desalination and district heating (2). Their front-end fuel cycle and back-end waste management can have ecological implications that need to be explored.

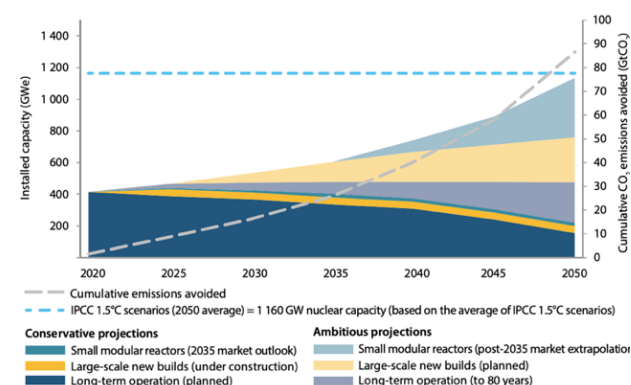


Fig. 1: NEA net zero projection for global installed nuclear capacities by 2050 (1)

2. Sustainability framework

Sustainability has become one of the main drivers for investment, where greenhouse gas (GHG) emissions are a primary indicator.

Although lifecycle emissions occur from fossil-fuel reliant construction, uranium mining and milling, operation and plant decommissioning, nuclear power has the potential to avert a cumulative 90 Gt of CO₂ emissions by 2050, as shown in Fig. 1 (1,3). The mean lifecycle GHG emissions for LRs are around 66 g/CO₂-eq/kWh, a substantial drop from 443 g/CO₂-eq/kWh for combined cycle natural gas (3). Yet, a Westinghouse SMR can achieve 9.1 g/CO₂-eq/kWh, attributed to efficiencies in construction, operation & maintenance and decommissioning (4).

Although SMRs can provide material reductions in carbon footprint compared to LRs and natural gas, it is critical not to become complacent. GHG emission estimates for SMR often overlook its complex waste streams and increasing use of high-assay, low-enriched uranium (HALEU). The environmental implications of the front-end fuel cycle and back-end waste management require investigation and evaluation for the sustainable deployment of SMR.

3. From conventional LEU to innovative HALEU

Higher fuel enrichment can deliver better economic and operational value for the nuclear power industry. Compared to the 3-5% low-enriched uranium (LEU) that the majority of commercial LRs rely on, SMR technologies allow for HALEU. Enriched up to 20%

isotope content, certain SMR designs can more than triple burn-up, energy extracted per mass of fuel across lifespan, of current LR technologies using HALEU and innovative cladding materials (1). Higher burn-up reduces net fuel costs and extends cycle length, time interval for refuelling and maintenance (5). With more compact cores, these factors directly increase the capacity factor, ratio of actual energy outputted against theoretical over time, directly impacting economic efficiency. However, transitioning from traditional LEU to HALEU introduces supply chain and upstream challenges that have environmental repercussions.

3.1 Barriers to HALEU adoption

HALEU is difficult to obtain, with commercial production limited to Russia. Its constrained supply is exacerbated by current geopolitical uncertainties, threatening disruptions to the international nuclear fuel supply chain. To reduce dependency on nuclear exports and proliferation risk during transportation, the US, China and UK are developing the capabilities to produce HALEU domestically (1).

HALEU requires five times the natural uranium compared to LEU (1). Meeting this increased demand would require constructing or expanding enrichment facilities, with the corresponding energy consumption, industrial emissions and land use. This would effectively raise fuel cycle emissions for SMR, which constitute 51% of lifecycle emissions (4), by more than a factor of five. The significant increase in natural uranium feedstock would exert pressure on the upstream uranium mining industry. Being carbon- and water-intensive, the unregulated oversight of the uranium mining industry has a historic reputation for soil, food chain and groundwater contamination, leading to environmental and public health complications in its communities (6).

Constrained supply chains, geopolitical risks, and limited industry experience hinder the adoption of HALEU. Although HALEU can provide the economic and operational value proposition for some SMR concepts, its commercialisation is accompanied by lifecycle emissions, environmental and social impacts from the upstream uranium mining and enrichment industry.

4. Complicated waste streams

The nuclear waste stream, made from spent fuel, fission products and coolant, differs across reactor designs in terms of volume, composition, stability and disposal requirements. The smaller dimensions of the SMR should produce less waste compared to LRs. However, on an energy equivalent volume basis, molten salt and sodium-cooled SMR concepts can increase the volume of nuclear waste required for disposal by factors of up to 30, which can be attributed to the use of neutron reflectors, chemically reactive fuels and coolants (7).

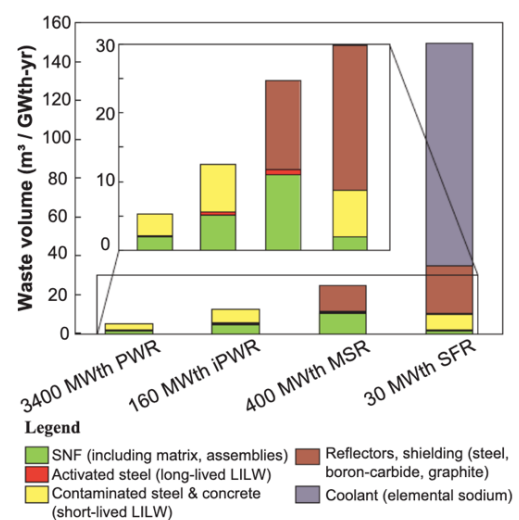


Fig. 2: Energy-equivalent volumes, by waste type, for various SMR designs (7)

Waste volume should not be used as the primary evaluation metric. Instead, waste streams are categorised into low-, intermediate- or high-level waste (HLW), depending on radioactivity and decay heat rate. Fissile, radiological, physical and chemical properties determine the complexity, cost and safety of managing the waste stream (7). Even at low levels, radiation is extremely hazardous to the health of humans and other living organisms, associated with acute radiation syndrome, increased cancer risk and long-term physical health, social and psychological problems.

Disposal methods range from storage at near-surface facilities for low-level waste to long-term storage in deep geological repositories for HLW, through vitrification or solidifying into a glass-like matrix (8). LRs generate stable low- and intermediate-level

waste, which contains nominal forms of concrete, steel and zirconium cladding and can be stored at ambient, near-surface facilities (7). Conversely, waste streams from SMR contain chemically unstable coolants, such as molten salts and liquid metals, which are susceptible to exothermic reactions or nuclear criticality upon contact with water and other repository materials (8). These require treatment and conditioning before disposal, generating secondary waste streams that further complicate waste management.

More HLW, high fissile nuclide concentration, is generated from the enhanced neutron leakage escaping the core, which induces radioactivity in the surrounding equipment (7). These heterogeneous waste streams may be incompatible with current disposal strategies, such as dry cask storage and geological disposal, and increase the fuel cycle cost, radiation exposure and proliferation risks (8).

4.1 Redefining waste management

Some emerging SMRs can alleviate the nuclear waste disposal problem by burning waste actinides and excess fissile nuclides such as plutonium and uranium still present in spent fuel (2). This not only increases fuel utilisation by maximising the amount of energy extracted per unit mass, but it can also reduce the volume of HLW in waste streams. Although these types of reactors can play an essential role in the waste management process, this could further complicate the fuel cycle.

However, this does not solve the complex waste issue. Operational variables, reactor design and waste properties from SMR plants require careful evaluation before deployment. Widespread deployment without effective back-end management can generate long-term waste issues and compromise the environmental benefits of nuclear power. A realistic environmental analysis should not only consider the volume of nuclear waste in its lifecycle assessment, but also the complexity, treatability and resources required for disposal.

5. Future steps for sustainable deployment

Technology for LEU fuel cycles and LR waste management is established in the nuclear industry;

however, the introduction of HALEU and SMR technology complicates the front-end fuel cycle and back-end waste management. The first hurdle is the limited industry experience and lack of a reliable supply chain for HALEU. Further research will be required to understand the potential implications of its entire fuel cycle, from upstream uranium mining to downstream waste characteristics. Coordinate efforts, through inter-governmental and public-private partnerships, will be necessary to harmonise regulatory frameworks, develop fuel cycle infrastructure and establish best practices for sustainable HALEU production and utilisation.

Effective waste strategies from treatment to storage need to be developed according to the distinct waste characteristics of SMR models to ensure safe and sustainable waste management. New planning efforts and centralised facilities will be needed to anticipate the diversity of waste characteristics and geographic decentralisation of SMR sites, such as an integrated waste management system tailored to distinct waste types. These reprocessing facilities could consolidate the waste management process, providing several stages of treatment, recovery of usable fissile material and vitrification for long-term disposal. This strategy has been adopted in France, Japan and Russia, but is not yet compatible with SMR waste characteristics (9). Transparent life cycle accounting will be necessary to evaluate the environmental effects of the front-end fuel cycle and back-end waste management process.

6. Conclusion

SMR can be a powerful tool in decarbonising the energy grid and many industries, offering lower upfront costs, flexibility and improved fuel utilisation. The increasing use of HALEU fuel and the complex waste characteristics of SMR technology pose technical, regulatory and environmental challenges. These are primarily caused by limited industry experience, a lack of reliable HALEU supply chain and incompatible waste strategies. Addressing these limitations through establishing governmental partnerships, a robust regulatory framework and front-end fuel cycle and back-end waste management infrastructure is key to optimising SMR for its role in a sustainable future.

KEY TAKEAWAYS

- SMR has an important role to play in decarbonising a variety of industries, from electricity generation to water desalination
 - SMRs can provide GHG emission reductions; however, highly enriched fuels and complex waste characteristics have environmental implications
 - Increasing use of HALEU can induce lifecycle emissions and supply chain challenges in the uranium mining and enrichment industry
 - SMR waste streams are incompatible with current disposal methods, requiring prior treatment
 - Addressing these challenges requires establishing governmental partnerships, a regulatory framework and fuel cycle and waste disposal infrastructure
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I acknowledge that this work is my own, and I used Copilot GPT (Microsoft 365, <https://copilot.microsoft.com>) to conduct initial research